

Ana Sofia Arango Yanza

Full-Stack Developer · AI & Agents

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SUMMARY

Systems Engineering student (9th semester, GPA 4.5/5.0) and full-stack developer with experience leading software projects for public-sector clients and building generative-AI solutions with LLMs and multi-agent systems. Passionate about detail, product design, and the intersection of software and artificial intelligence. Bilingual (Spanish/English).

EDUCATION

Universidad del Cauca

B.Sc. in Systems Engineering — GPA 4.5/5.0

Aug. 2022 – Jun. 2027 (expected)

Popayán, Colombia

EXPERIENCE

Team Lead & Full-Stack Developer

2026

Cauca Government Secretariat — Vigía Cauca Project

Popayán, Colombia

- Led the development team of a geo-referenced security-incident management platform, from requirements gathering to formal client handover.
- Built the frontend (React + Leaflet) and backend (Spring Boot + PostgreSQL) under a hexagonal, microservices architecture.
- Implemented role-based authentication, interactive dashboards with heat maps and exportable reports (PDF/Excel), with CI/CD on GitHub Actions.

Research Assistant — Data Science Research Group

2025 – Present

Universidad del Cauca

Popayán, Colombia

- Applied-AI research; **first place** at the Academic Projects Fair with a deep-learning model for dementia-level detection on MRI images.
- Two conference talks (dementia detection and Bayesian optimization of a CNN for P-wave detection).
- Active member of the IEEE Student Branch; support in organizing academic and technical events.

PROJECTS

AIMO — Python, Flask, React, LLMs, AWS Bedrock

2026

- Conversational emotional-support assistant for students with a cascading multi-agent system (LLaMA 3.3-70B, GPT-4, Claude Sonnet 4.5) and an LLM-as-a-Judge methodology to assess empathy in real time.
- Flask backend integrating Groq, OpenAI and AWS Bedrock; React frontend. Deployed to production (Vercel + Railway).

Dementia Detection · MRI — Python, CNN, Deep Learning

2026

- Dementia-level classification on MRI scans using convolutional neural networks. **First place** at Universidad del Cauca's Academic Projects Fair.

DevDynamics — Python, Mesa, React, Flask

2025

- Agent-based simulator of software team dynamics; models Brooks's Law, developer burnout and how technical debt affects team velocity.

TECHNICAL SKILLS

Languages:	Java, Python, TypeScript, JavaScript, C#, SQL, HTML/CSS
Frontend:	React, Next.js, Angular, React Native, Tailwind CSS, Leaflet
Backend:	Spring Boot, Flask, Node.js, ASP.NET Core, REST APIs
AI / Data:	LLMs, multi-agent systems, prompt engineering, TensorFlow, Scikit-learn, AWS Bedrock
Databases:	PostgreSQL, MySQL, MongoDB, Oracle
Tools:	Git, GitHub Actions, Docker, AWS, Figma · microservices / hexagonal architecture

CERTIFICATIONS & AWARDS

NVIDIA DLI: Rapid Application Development with LLMs · Building LLM Applications with Prompt Engineering · Generative AI Explained · Transformer-Based NLP · Getting Started with Deep Learning · Accelerating End-to-End Data Science Workflows

Cisco Academy: CCNA: Introduction to Networks · CCNA: Switching, Routing & Wireless Essentials · Junior Cybersecurity Analyst · Introduction to Cybersecurity

Awards: Dean's List Scholarship — Matrícula de Honor (3 times) · First place — Academic Projects Fair, Universidad del Cauca (2026)

LANGUAGES

Spanish: Native · **English:** Intermediate (B1-B2) — fluent technical reading and writing